EXPERIMENT-2

Aim:- Convert the ER diagram into a relational Schema. Write complete MySQL Create Table statements with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASADE and SET NULL constraints. Insert sample data and demonstrate referential integrity violations.

Source Code:-

```
CREATE DATABASE e_commerce;
USE e_commerce;

--Create Customer table
CREATE TABLE Customer(
        CustomerID INT PRIMARY KEY,
        firstName VARCHAR(50) NOT NULL,
        MiddleName VARCHAR(50),
        LASTName VARCHAR(50));

--Multiple email address of customers
CREATE TABLE Customer_Email(
        CustomerID INT NOT NULL,
        Email VARCHAR(100) NOT NULL UNIQUE,
        PRIMARY KEY (CustomerID, Email),
        FOREIGN KEY (CustomerID)
            REFERENCES Customer(CustomerID)
                ON DELETE CASCADE
        );
```

```
--Multiple phone numbers of customers
CREATE TABLE Customer_Phone (
    CustomerID INT NOT NULL,
    PhoneNo VARCHAR(15) NOT NULL UNIQUE,
    PRIMARY KEY (CustomerID, PhoneNo),
    FOREIGN KEY (CustomerID)
        REFERENCES Customer(CustomerID)
        ON DELETE CASCADE
);

--Create Address table
CREATE TABLE Address(
    AddressNo INT PRIMARY KEY,
    CustomerID INT NOT NULL,
    HouseNo VARCHAR(50) NOT NULL,
    City VARCHAR(50) NOT NULL,
    State VARCHAR(50) NOT NULL,
    PINCode INT NOT NULL,

    PRIMARY KEY (CustomerID, HouseNo),
    FOREIGN KEY (CustomerID)
        REFERENCES Customer(CustomerID)
        ON DELETE CASCADE

--Multiple landmarks for an address/House no.
CREATE TABLE Address_landmark (
    HouseNo VARCHAR(50) NOT NULL,
    Landmark VARCHAR(150) NOT NULL,
    PRIMARY KEY(HouseNo, Landmark),
    FOREIGN KEY(HouseNo)
        REFERENCES Address(HouseNo)
        ON DELETE CASCADE
);
```

```
-- Create Order Table
CREATE TABLE Orders(
    OrderID INT PRIMARY KEY,
    CustomerID INT NOT NULL,
    HouseNo VARCHAR(50) NULL,
    OrderDate DATE NOT NULL,
    Status VARCHAR(30) NOT NULL,

    FOREIGN Key (CustomerID)
        REFERENCES Customer(CustomerID)
        ON DELETE CASCADE,

    FOREIGN KEY (HouseNo)
        REFERENCES Address(HouseNo)
        ON DELETE SET NULL
);

-- Create Payment Table
CREATE TABLE Payment(
    PaymentID INT PRIMARY KEY,
    OrderID INT NOT NULL UNIQUE,
    Amount DECIMAL(10,2) NOT NULL,
    PaymentDate DATE NOT NULL,
    Method VARCHAR(30) NOT NULL,
    FOREIGN KEY(OrderID)
        REFERENCES orders(OrderID)
        ON DELETE CASCADE,
);
```

```
-- Create Delivary table
CREATE TABLE Delivary (
        DelivaryID INT PRIMARY KEY,
        OrderID INT NOT NULL UNIQUE,
        HouseNo INT NULL,
        DelivaryDate DATE,
        DelivaryStatus VARCHAR(30) NOT NULL,

      FORIEGN KEY (OrderID)
        REFERENCES Orders(OrderID)
        ON DELETE CASCADE,

      FORIEGN KEY (HouseNo)
        REFERENCES Address(HouseNo)
        ON DELETE SET NULL.
    );

-- Create Category table
CREATE TABLE Category (
        CategoryID INT PRIMARY KEY,
        CategoryName VARCHAR(100) NOT NULL UNIQUE,
      Description VARCHAR(255)
    );

-- Create Product table
CREATE TABLE Product (
      ProductID INT PRIMARY KEY,
    CategoryID INT Null,
    Name VARCHAR(100) NOT NULL,
    Price DECIMAL(10,2) NOT NULL,
    Stock INT NOT NULL,
```

```
    FOREIGN KEY (CategoryID)
        REFERENCES Category (CategoryID)
        ON DELETE SET NULL
    );

-- Create Order_Item table
    CREATE TABLE Order_Item (
        OrderID INT NOT NULL,
        ProductID INT NOT NULL,
        QUANTITY INT NOT NULL,
        UnitPrice Decimal (10,2) NOT NULL,
        PRIMARY KEY (orderID, ProductID),
        FOREIGN KEY (OrderID)
            REFERENCES orders (OrderID)
            ON DELETE CASCADE,

        FOREIGN KEY (ProductID).
            REFERENCES Product (ProductID)
            ON DELETE CASCADE,
    );

-- Create VIEW Order_Item_View
    CREATE VIEW Order_Item_View AS
    SELECT
        OrderID, ProductID, QUANTITY,
        UnitPrice,
        UnitPrice * 0.05 AS Discount
    FROM Order_Item;
```

```
--Create Seller Table
CREATE TABLE Seller(
        SellerID INT PRIMARY KEY,
        firstName VARCHAR(50) NOT NULL,
        MiddleName VARCHAR(50),
        LastName VaRCHAR(50) NOT NULL,
        Phone  VARLHAR(20) NOT NULL UNIQUE
    );


-- Create Seller email address -
 CREATE  TABLE Seller_Email(
        SellerID INT NOT NULL,
        Email VARCHAR(100) NOT NULL UNIQUE,
        PRIMARY KEY (SellerID, Email),
        FOREIGN KEY (SellerID)
             REFERENCES Seller(SellerID)
             ON DELETE CASCADE
         );

-- Create seller_Category
   CREATE TABLE Seller-Category(
      SellerID INT NOT NULL,
      CategoryID INT NOT NULL,
     PRIMARY KEY (SellerID, CategoryID),
     FOREIGN KEY (SellerID)
          REFERENCES seller(SellerID)
          ON DELETE CASCADE

     FOREIGN KEY (categoryID)
          REFERENCES category(categoryID)
          ON DELETE CASCADE

    );
```

```
--Inserting in tables
INSERT INTO Customer
(CustomerID), firstName , MiddleName, LastName )
VALUES
(1, 'Aisha', 'K', 'Sharma'),
(2, 'Rahul', Null, 'Verma'),
(3, 'Neha', 'R', 'Singh');

-- Insert customer_Email
InSert INTO Customer_Email
(CustomerID, Email) Values
(1, 'aisha@gmail.com'),
(2, 'rahul@gmail.com'),
(3, 'neha@gmail.com');

--Insert Phone number of customer
Insert Into Customer_Phone
(CustomerID , PhoneNO) Values
(1, '9876543210');
(2, '9876543211');
(3, '9876543212');

--Insert address
Insert Into Address
(AddressID , CustomerID, HouseNo, City, State, PINCode)
VALUES
(101 , 1, '12A', 'Kanpur', 'Uttar Pradesh', '208001'),
(102 , 2, '45B', 'Delhi', 'Delhi', '110001'),
(103 , 3, '78C', 'Lucknow', 'Uttar Pradesh', 226001');

-- FOR Integrity
Insert Into Orders
(orderID, CustomerID, AddressID, orderDate, Status) Values
(404, 999, 101, '2026-10-03', 'Placed');

Insert Into order_Item
(orderID, ProductID, Quantity, unitPrice, Discount) Values
(401, 999, 1, 1000.00, 0.00);
```

```
Select * from Customer;
Select * from Product;
Select * from Orders;
Select * from Order_Item;
```

Output -

• Customer Table -

CustomerID	firstName	MiddleName	LastName
1	Aisha	K	Sharma
2	Rahul	Null	Verma
3	Neha	R	Singh

• Product Table -

ProductID	CategoryID	Name	Price	Stock
201	1	Laptop	65000.00	10
202	1	Headphones	2500.00	25
203	2	Jacket	3000.00	15

• Order Item Table -

OrderID	ProductID	Quantity	UnitPrice	Discount
401	201	1	65000.00	5.00
401	202	2	2500.00	0.00
402	203	1	3000.00	10.00
403	204	2	800.00	5.00

- Referential Integrity Violation 1

```
Insert INTO orders
(OrderID, CustomerID, AddressID, OrderDate, Status) VALUES
(4043, 999, 101, '2026-10-03', 'Placed');
```

Result:-

Error 1452 (23000): Cannot add or update a row: a foreign key constraint fails.

- Referential Integrity Violation 2

for the invalid ProductID:

Error : Cannot add or update a row: constraint fails.

- Referential Integrity Violation 3: Inserting a duplicate primary key:-

```
INSERT INTO Customer
(CustomerID, firstName, MiddleName, LastName) VALUES
(1, 'Nail', Null, 'Sharma');
```

Output -

Duplicate entry error (ERROR 1062).

Reason: CustomerID 1 already exists in the Customer table.

A primary key must be unique.

- Inserting NULL into a NOT NULL column

Output:- Column 'firstName' cannot be null (typically Error 1048).

- Deleting a parent record referenced by a foreign key

Output - This deletion should succeed, because your Seller_Category table uses on Delete CASCADE. The associated seller-category record will be deleted automatically.